

Minwoo Park

mwpark@engineering.upenn.edu

Education

University of Pennsylvania

Expected December 2026

Master of Science — Electrical Engineering, Concentration in AI

G.P.A.: 3.62

California Polytechnic State University, San Luis Obispo

Sep 2021 – Jun 2025

Bachelor of Science — Electrical Engineering, Minor in Computer Science

G.P.A.: 3.959, Dean's List: Fall 2021 – Spring 2025, and President's Honor List: 2021 – 2025

Skills

Technical Skills: Altium, Eagle, Vivado, LTspice, SPICE, STM32, Arduino, FPGA, Fusion 360, Firebase, AWS, MS Azure, MS Power Automate, I2C, UART, SPI, Oscilloscope, Spectrum Analyzer, Network Analyzer, Multimeter

Interpersonal Skills: Problem-solving, Creativity, Adaptability, Conflict resolution, Time management, Teamwork

Programming: C, C++, Java, Python, JavaScript, SQL, SystemVerilog, RISC-V Assembly, MATLAB, Simulink, LabVIEW, ROS

Languages: English, Korean, Spanish (Limited)

Work Experience

Varian (Siemens Healthineers) — Palo Alto, CA

May 2026 – Present

Embedded Software Engineer/ Control Systems Intern

- Implemented control and optimization algorithms to improve servoing performance and motion accuracy in radiotherapy machine systems.
- Developed and optimized path and treatment planning algorithms for radiotherapy machine trajectories, improving planning efficiency and system performance.

Kumar Lab — Philadelphia, PA

Aug 2025 – Present

Research Assistant (avg. 10 hrs/week)

- Developing a research paper on multi-agent payload control, including literature synthesis, problem formulation, and analysis of trajectory generation and controller performance.
- Built and configured a Crazyflie experimental testing framework for evaluating trajectories and controllers, supporting reproducible validation of aerial robotics experiments.

CPSquare (Cal Poly Cyber-Physical Systems Lab) — San Luis Obispo, CA

Sep 2024 – Jun 2025

Research Assistant (avg. 10 hrs/week)

- Designed and implemented dynamic formation control algorithms enabling multi-drone coordination and real-time failure recovery.
- Developed Kalman filter localization algorithms for a swarm of Crazyflie quadcopters, optimizing performance under hardware and sensor constraints.
- Built failure detection and recovery mechanisms ensuring robustness under variable environmental and sensor conditions.

Cal Poly Dept. of Electrical Engineering — San Luis Obispo, CA

Jan 2025 – Jun 2025

Instructional Teaching Assistant (avg. 15 hrs/week)

- Graded assignments and provided feedback for EE 302 (Classical Controls) and EE 342 (Classical Control Systems Lab).

Stryker – Endoscopy — San Jose, CA

Jun 2024 – Sep 2024

Advanced Operations Electrical Engineering Intern

- Introduced new automation software for the 1788 Camera Head manufacturing line using LabVIEW, reducing manufacturing time by up to 20%.
- Revitalized Precision Light Source calibration software for compatibility with the L11 Light Source line, saving 5% in production floor space.
- Revised 1788 Camera Head quality check software, increasing reliability and reducing debugging time by 2+ weeks.

Cal Poly Dept. of Electrical Engineering — San Luis Obispo, CA

Oct 2023 – Jun 2024

Instructional Teaching Assistant (avg. 10 hrs/week)

- Taught soldering, PCB design, and microcontroller fundamentals to EE 143 (Circuit Analysis Lab) students.
- Graded assignments for EE 113, EE 228, and EE 342.

Stryker – Interventional Spine — Portage, MI

Jun 2023 – Sep 2023

R&D Electrical Engineering Intern

- Developed a PCBA in Altium and integrated it with Arduino, reducing MultiGen 2 splitter box testing time by 75%.
- Tested and verified EMC and power compliance for RF ablation console PCBAs following IEC 60601 standards.
- Authored comprehensive documentation for the MultiGen 2 testing device for R&D teams.

Cal Poly Corporation — Corporate Engagement & Innovation — San Luis Obispo, CA Aug 2022 – Jul 2023

Student Assistant (avg. 10 hrs/week)

- Built MS Power Automate workflows and Python scripts to automate research grant/contract data collection.
- Developed a CRM application using JavaScript and Firebase.

Code Ninjas Cupertino — Cupertino, CA

Sep 2019 – Aug 2021

Code Sensei / Instructor

- Taught programming fundamentals and intermediate concepts through JavaScript.
- Guided students through debugging and development of multiple games.

Projects

Crazyflie Drone Swarm Controller

Sep 2024 – Jun 2025

- Built a scalable swarm controller using real-time peer-to-peer communication and distributed algorithms to coordinate synchronized multi-drone flight.

Vision Controlled Robot Manipulator

Sep 2024 – Dec 2024

- Implemented forward/inverse kinematics and computer vision algorithms to control an OpenManipulator-X robot arm for pick-and-place operations.

Ultrasonic Underwater Sensor Network

Sep 2024 – Dec 2024

- Built an end-to-end network transmitting DHT11 sensor data through ultrasonic hardware, AGC, PLL tone decoder, and a Raspberry Pi server for web display.

Custom IR RC Car

Apr 2024 – Jun 2024

- Integrated a Fusion 360–designed 3D chassis with LCD, accelerometer, IR sensors, ultrasonic sensors, and motor drivers to build an IR-controlled RC car.

RISC-V 32-bit MCU

Jan 2024 – Mar 2024

- Implemented a 32-bit pipelined RISC-V CPU with cache and peripheral I/O using SystemVerilog on a Basys-3 FPGA.

4-Bit DAC PCBA

Apr 2022 – Jun 2022

- Designed a 4-bit DAC with binary-weighted resistor architecture and integrated it with an Arduino speaker system.

Relevant Coursework

Electrical / Computer Engineering: Circuit Analysis I/II/III, Digital & Computer Design, Assembly Programming, Digital & Continuous Systems, Classical Controls, Digital & Analog Electronics, Microcontroller System Design, Vision-Based Robotics, Optimization Theory, Modern Convex Optimization, Learning in Robotics, Embedded Systems/Devices, and Linear Systems Theory

Computer Science: Data Structures, Object-Oriented Programming/Design, Discrete Structures, Systems Programming, Machine Learning, Machine Learning for Time-Series Data, Principles of Deep Learning, Big Data Analytics, and Statistics for Data Science

Certifications

FCC Amateur Radio Technician License